

Fault Location Using Impedance-Based Algorithms on Non-Homogeneous Feeders

S. Kulkarni, N. Karnik, S. Das and S. Santoso

Abstract—Impedance-based algorithms like the positive-sequence reactance and Takagi methods are derived assuming a homogenous line conductor. This assumption is violated in practical distribution feeder circuits. The objective of this paper is to demonstrate that these methods can still be effectively applied to non-homogenous feeders by using the line parameters of the most commonly occurring conductor type in the circuit. This approach is first tested on a modified IEEE Test Feeder by simulating two cases, namely, the reference case and the case with a homogenous feeder. For the positive-sequence reactance method the maximum absolute error in both cases is about 8%, while for the Takagi method it is 6.92% and 9.53% in the homogenous case and reference case, respectively. The proposed approach is then demonstrated using ten fault cases on utility distribution feeders. The average absolute error obtained for the positive-sequence reactance and Takagi estimates is 10.23% and 9.34%, respectively, which corresponds to a median error value of 0.16 miles and 0.15 miles in the location estimates.

Index Terms—Diagnosis (fault), power distribution faults, power system monitoring

I. INTRODUCTION

Impedance-based algorithms are widely used for fault location on utility distribution feeders [1]–[4]. This paper focuses on two of the most popular algorithms, namely, the positive-sequence reactance and Takagi methods [5]. These methods are derived assuming system homogeneity or uniformity. In other words, the conductor is assumed to have constant line impedance parameters throughout the circuit. However, this assumption is violated in practical utility distribution feeders which are constructed using different conductor types and pole configurations. The positive- and zero-sequence impedances of the conductors vary according to the configurations. For example, a utility circuit analyzed later in this paper is constructed using three different wire types. 0.13 miles of the circuit uses 2-ACSR wire type for all three phases, 1.25 miles uses 336-AAC and 6.23 miles uses 556-AAC. The neutral conductor wire in these three cases is of type 2-ACSR, 1/0-ACSR and 336-AAC, respectively. Such a circuit is considered as a non-homogenous distribution feeder. The objective of this paper is to demonstrate that the impedance-based methods can be successfully applied to locate faults on non-homogenous feeders while preserving the accuracy level.

The approach taken in this paper is to assume the feeder to be constructed using the most commonly occurring conductor

in the circuit. The first step in this process is to generate a list of all the conductor types used in the circuit along with their section length. In the next step, the most commonly occurring conductor type is to be chosen and its positive- and zero-sequence line parameters are to be employed in the impedance-based fault location methods. This proposed approach is first tested on a modified IEEE 34 Node Test Feeder modeled in PSCAD. Faults are placed at different locations along the feeder and the reactance to the fault is estimated using the positive-sequence reactance and Takagi methods. The line parameters of the most commonly occurring conductor are utilized in these algorithms. To study the effect of system non-homogeneity, a new homogeneous test feeder model is simulated where the entire circuit is constructed using the most commonly occurring line configuration in the reference case. Faults are placed at the same locations and the positive-sequence reactance and Takagi methods are again applied. The error percentages in both the cases are comparable. For the positive-sequence reactance method, the maximum absolute error in both cases is about 8%. In contrast, the maximum absolute error for the Takagi method is 6.92% and 9.53% in the homogenous case and reference case, respectively. The proposed approach is then demonstrated using ten fault cases on utility distribution feeders. The average absolute error obtained for the positive-sequence reactance and Takagi estimates is 10.23% and 9.34% respectively. Correspondingly, the median value of the absolute difference between the estimate and the actual fault distance is 0.16 miles and 0.15 miles, for the positive-sequence reactance and Takagi methods, respectively.

An overview of the positive-sequence reactance and Takagi methods is provided in Section II. Modified IEEE Test Feeder is described in Section III. Section IV details the fault location on the reference case, while the case with the homogenous feeder is explained in Section V. Demonstration of the proposed approach using faults on utility distribution feeders is presented in Section VI with the help of ten examples cases.

II. REVIEW OF IMPEDANCE-BASED METHODS

Impedance-based methods are commonly employed for fault location in utility distribution feeders [2]. These methods require the fault voltage and current phasor quantities. Additionally, most methods require the positive- and zero-sequence line impedances, z_1 and z_0 (in Ω/mile) as well as prefault load currents [5]. The positive-sequence reactance and Takagi methods are two of the most common impedance-

S. Kulkarni, N. Karnik, S. Das and S. Santoso are with The Department of Electrical and Computer Engineering at The University of Texas at Austin.

based methods used to locate single line-to-ground faults. These methods are briefly described in this Section with the help of Fig. 1.

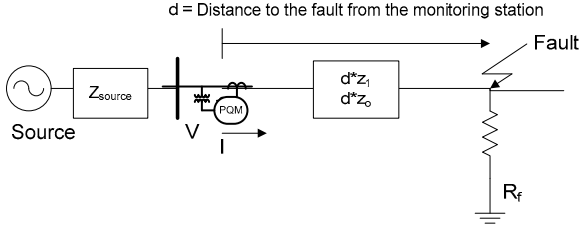


Fig. 1. Single Line-to-ground Fault on a Distribution Feeder Under Uniform Line Assumption

A. Positive-sequence Reactance Method

The positive-sequence reactance method is one of the simplest as well as efficient methods used for fault location [3] [4]. Fault location estimates are given in terms of the positive-sequence reactance and can be converted to actual distance. This method computes the reactance to the fault (as given by (1)) in order to overcome the effect of fault resistance.

$$X_1 = \text{imag} \left(\frac{V}{I_S} \right)$$

$$I_S = I + kI_0 \quad (1)$$

$$k = \frac{z_0 - z_1}{z_1}$$

Where,

V = Faulted phase voltage at monitoring location

I = Faulted phase current at monitoring location

I_0 = Zero-sequence component of faulted phase current

z_1 = Positive-sequence impedance of distribution line per unit length

z_0 = Zero-sequence impedance of distribution line per unit length

X_1 = Positive-sequence reactance to the fault measured from monitoring location

B. Takagi Method

The Takagi method [6] requires both the prefault and fault current data. This method takes into account the load current and minimizes the effect of fault resistance. The distance is estimated using (2).

$$d = \frac{\text{Im}(V I_{\text{sup}}^*)}{\text{Im}(z_1 I_S I_{\text{sup}}^*)} \quad (2)$$

$$I_{\text{sup}} = I - I_{\text{preflt}}$$

Where,

I_{preflt} = Prefault load current in faulted phase

I_{sup} = Superposition current

Both the positive-sequence reactance and Takagi methods make the assumption that the distribution feeder under consideration has a uniform or homogenous line configuration. This means that the positive- and zero-sequence line parameters z_1 and z_0 per unit length of the line conductor are the constant for the entire feeder length.

III. DESCRIPTION OF MODIFIED IEEE TEST FEEDER

A modified IEEE 34 Node Test Feeder shown in Fig. 2 has

been used in this paper [7]. This is an actual feeder located in Arizona. It has multiple single-phase laterals, three-phase branches and a wide variety of components which makes it an ideal test feeder. This test feeder is modeled in the time-domain using PSCAD/EMTDC simulation software [8].

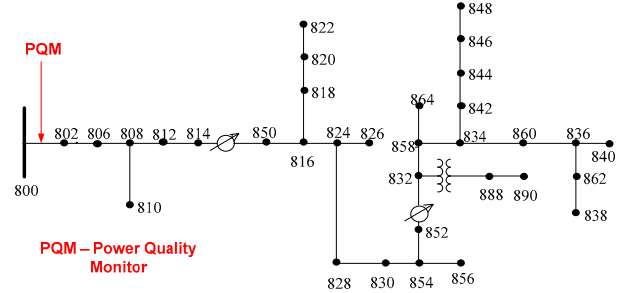


Fig. 2. One-Line Diagram of the Modified IEEE 34 Node Test Feeder

The load current in this test feeder was around 40A (rms) and the fault current at the farthest laterals was around 300A (peak). Such low levels of load current are not useful for fault location analysis since the location estimates are hardly affected and no effective conclusions can be made. Hence, some modifications were implemented to make the test circuit more useful for analysis. The length of the main feeder was decreased from 35 miles to 9.54 miles. The lengths of the single-phase laterals were reduced from about 8-9 miles to 1-3 miles. The level of fault currents in the system was considerably increased from 5kA (for faults closer to substation) to 970A (for faults at maximum distance from substation). Furthermore, the system load was also increased from 40A/phase to 425A/phase. Some features of this modified IEEE 34 Node Test Feeder are given below:

- Nominal feeder voltage is 24.9 kV
- Single-phase laterals at nodes 808 to 810, 816 to 822, 824 to 826, 854 to 856, 858 to 864 and 836 to 838
- Three-phase laterals at nodes 832 to 890 and 834 to 848

The test feeder uses five line configurations, namely, 300, 301, 302, 303 and 304. Line configurations 300 and 301 are three-phase while, 302, 303 and 304 are single-phase. The positive- and zero-sequence impedances and the segment lengths of these configurations are given in Table I. Note that the length of the main feeder is 9.54 miles, while the total conductor length, including all the laterals is 19.11 miles.

TABLE I
LINE CONFIGURATION DATA FOR MODIFIED IEEE 34 NODE TEST FEEDER

Parameter	300	301	302	303	304
z_0 (Ω /mile)	1.749 + $j2.372$	2.387 + $j2.578$	0.9332 + $j0.4952$	0.9332 + $j0.4952$	0.6406 + $j0.4737$
z_1 (Ω /mile)	1.120 + $j0.833$	1.690 + $j0.841$	0.9332 + $j0.4952$	0.9332 + $j0.4952$	0.6406 + $j0.4737$
length (mile)	3.30	9.60	2.32	2.97	0.92

IV. REFERENCE CASE FOR FAULT LOCATION

The modified IEEE Test Feeder described in Section III is used here to establish a reference case. The single line diagram of this feeder is shown in Fig. 3. A power quality monitor at

node 800 samples the voltage and current waveforms at a rate of 32 samples per cycle. Eight bolted (i.e. $R_f = 0\Omega$) single line-to-ground faults are placed at different locations along the test feeder. The faults are on nodes 808, 814, 830, 832, 840, 844, 848 and 860.

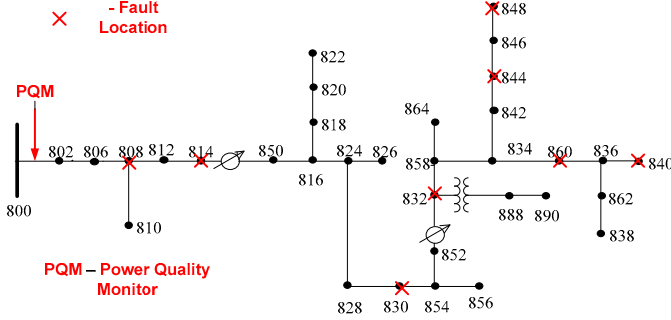


Fig. 3. Fault locations along the Modified IEEE 34 Node Test Feeder

The positive-sequence reactance and Takagi methods are used to estimate the reactance to the fault for all eight cases. Estimation errors are computed for each case using (3).

$$\text{Error(\%)} = \frac{\text{Estimated reactance} - \text{Actual reactance}}{\text{Actual reactance}} \times 100 \quad (3)$$

It was stated in Section II that the positive-sequence reactance and Takagi algorithms are derived assuming a homogenous distribution feeder. On the contrary, the modified IEEE 34 Node Test Feeder has five line configurations as summarized in Table I. Out of the total conductor length of 19.11 miles, configuration 301 is used for 9.60 miles of the feeder. In this paper it is proposed to utilize the sequence parameters of the most widely used line configuration in fault location algorithms. Hence we use the positive- and zero-sequence impedances corresponding to configuration 301 for the test feeder. Table II gives the reactance to the fault estimates for positive-sequence reactance and Takagi methods. Estimation error for each fault location is computed using (3).

TABLE II
REACTANCE TO THE FAULT ESTIMATES FOR REFERENCE CASE

Fault Node.	Actual +ve Seq. Reactance (Ω)	Positive-sequence Reactance Method		Takagi Method	
		Estimated +ve Seq. Reactance (Ω)	Error (%)	Estimated +ve Seq. Reactance (Ω)	Error (%)
814	2.230	2.262	1.42	2.081	-6.68
830	3.459	3.461	0.05	3.294	-4.77
860	5.872	5.692	-3.05	5.589	-4.81
848	7.992	7.303	-8.62	7.230	-9.53
808	1.197	1.219	1.80	1.123	-6.22
832	4.066	4.052	-0.34	3.893	-4.25
840	7.386	6.864	-7.07	6.813	-7.75
844	6.478	6.208	-4.16	6.109	-5.70

For a single-line to ground fault at node 814, the Takagi method gives an estimate of 2.081 Ω which is less than the actual fault reactance of 2.230 Ω . The estimation error computed using (3) is -6.68%. The negative sign indicates that

the Takagi method underestimates the fault location. On the other hand, for the same fault at node 814, the positive-sequence reactance method gives an estimate of 2.262 Ω . The estimation error is positive indicating an overestimation. From Table II, it is obvious that the Takagi method consistently underestimates the fault locations with estimation errors of -4.25% to -9.53%. Conversely, the positive-sequence reactance method has an estimation error ranging between +1.8% and -8.62%. Based on these results, it appears that the performance of these two algorithms is comparable; however, the positive-sequence reactance method tends to deliver more accurate location estimates. It should also be noted that performance of both the methods deteriorates as the distance of the fault from the monitoring location increases.

V. FAULT LOCATION USING HOMOGENOUS SYSTEM MODEL

The modified IEEE 34 Node Test Feeder used as a reference case in Section IV is non-homogeneous and has five different conductor configurations as given in Table I. Since configuration 301 is most widely used (9.60 miles out of 19.11 miles of total feeder length), the positive-sequence reactance and Takagi methods compute the reactance to the fault using values of z_1 and z_0 of this configuration. To study the effect of system non-homogeneity, a new homogeneous IEEE 34 Node Test Feeder model is created in PSCAD. All sections of the main feeder as well as the single- and three-phase laterals are constructed using configuration 301. As the length of the lines between adjacent nodes is unchanged, the distance to the fault measured from the monitoring location (node 800) is the same as the reference case. However, since the line configurations have changed the reactance to the fault measured from the monitoring location (node 800) changes.

Single line-to-ground faults are placed in the new homogeneous model at the same eight locations as the reference case shown in Fig. 3. Fault location is carried out using the positive-sequence reactance and Takagi methods as shown in Table III. The use of z_1 and z_0 values corresponding to configuration 301 for fault location is justified since this feeder is homogeneous. In contrast, for the reference case, where there are multiple conductor configurations, using values corresponding to a single configuration is erroneous. Note that the reactance to the fault measured from the monitoring location for the homogeneous feeder is different from the reference case. Hence the actual positive-sequence reactance to the fault (column 2 in Table III) has been recomputed for each location. This value is used to calculate estimation errors for the homogeneous feeder using (3). Table III shows that, the absolute value of the estimation error for the Takagi method has reduced for the homogenous case. Thus the accuracy of the reactance to the fault estimates has improved compared to the reference case. For a fault at node 860, when the system is non-homogenous (reference case), the Takagi method estimates the reactance to the fault with a -4.81% error. For the homogeneous system the error is -1.81%. Thus the assumption of system homogeneity for a non-homogeneous system increases the error percentage by 3%. For the positive-sequence reactance method, the error

percentages for the two cases are almost identical.

TABLE III
REACTANCE TO THE FAULT ESTIMATES FOR HOMOGENOUS FEEDER

Fault Node	Actual +ve Seq. Reactance (Ω)	Positive-sequence Reactance Method		Takagi Method	
		Estimated +ve Seq. Reactance (Ω)	Error (%)	Estimated +ve Seq. Reactance (Ω)	Error (%)
814	2.254	2.217	-1.66	2.235	-0.87
830	3.484	3.431	-1.54	3.463	-0.60
860	5.897	5.689	-3.52	5.790	-1.81
848	8.017	7.318	-8.71	7.462	-6.92
808	1.211	1.193	-1.50	1.201	-0.88
832	4.090	4.029	-1.50	4.072	-0.44
840	7.411	6.866	-7.36	7.028	-5.17
844	6.502	6.217	-4.40	6.325	-2.74

It can be concluded from the above analysis that the difference between error percentages in the reactance to the fault estimates for the homogeneous and non-homogeneous cases is marginal. Note that the objective of this paper as stated in Section I was to demonstrate that the impedance-based methods can be successfully applied to locate faults on non-homogenous feeders while preserving the accuracy level. Using the line parameters of the most commonly occurring line conductor for fault location meets this objective.

VI. APPLICATION TO FAULTS ON UTILITY DISTRIBUTION FEEDERS

In this Section the performance of the positive-sequence reactance and Takagi algorithms is evaluated using field data from ten utility cases. The voltage and current waveforms captured by SEL relays during the fault are made available for all events along with their circuit models. The circuit models are used to find the most commonly occurring three-phase conductor whose positive- and zero-sequence impedance parameters are used in distance estimation algorithms. The estimates are then compared to the actual fault locations to test the accuracy of the proposed approach. Two cases are explained in detail, followed by a summary of all ten cases.

A. Utility Test Case 1

The circuit diagram for utility test case 1 was constructed in CymeDist. The rated voltage at the substation is 34.5 kV while the short-circuit capacity calculated using the circuit model is 321MVA. A SEL-651R relay at the substation samples the three-phase line currents and line-to-neutral voltages at a rate of 32 samples per cycle.

On 9th March 2010 at 23:05, there was a lightning strike on the circuit which resulted in a momentary single line-to-ground fault on phase C. The relay recorded voltage and current waveforms and fault log are shown in Fig. 4. The relay recorded a prefault load current of about 40 A, while the maximum fault current magnitude recorded was 3.168 kA. The fault location was on a three-phase lateral 2.67 miles from the substation.

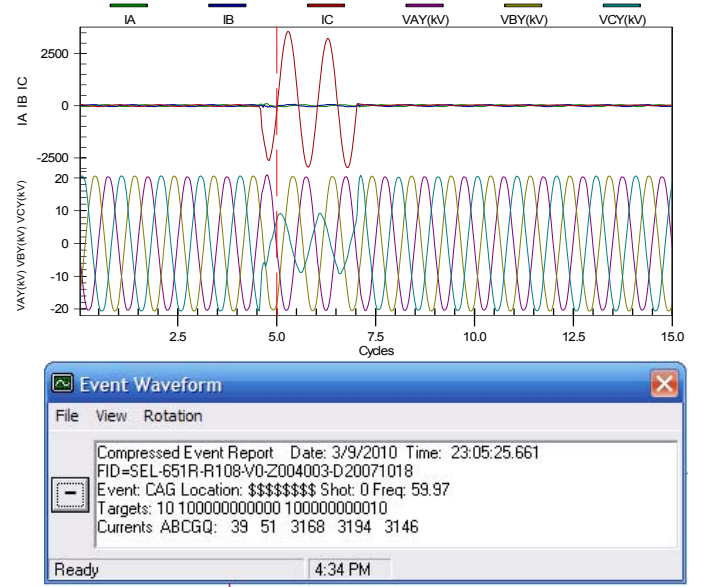


Fig. 4. Relay Data for Utility Test Case 1

The main feeder of this circuit is constructed using five different three-phase line configurations. The positive- and zero-sequence impedances and conductor lengths of these configurations are given in Table IV.

TABLE IV
LINE CONFIGURATION DATA FOR UTILITY TEST CASE 1

Parameter	2/0_ACSR, 2/0_ACSR, 2/0_ACSR, 2/0_A	2/0_ACSR, 2/0_ACSR, 2_ACS, 2_ACS	2_ACSR, 2_ACSR, 2_ACSR, 1/0_TPX	397_ACSR, 397_ACSR, 397_ACSR, 2/0_A	795_AAC, 795_AAC, 795_AAC, 397_ACSR
z_0 (Ω /mile)	1.1630+ j 2.3123	1.3316+ j 2.5572	0.8770 + j 2.1365	0.7100 + j 2.1244	0.3766 + j 1.8503
z_1 (Ω /mile)	0.6871+ j 0.7805	0.6871+ j 0.7806	0.6870 + j 0.7805	0.2341 + j 0.5926	0.1201+ j 0.5546
length (mile)	1.02	1.33	1.64	3.06	5.66
location	down- stream from fault	down- stream from fault	down- stream from fault	middle portion of main feeder either side of fault	close to sub- station along main feeder

The most commonly occurring conductor type in the circuit has the code, "795_ACC,795_ACC,795_ACC,397_ACSR". Using the proposed approach the positive- and zero-sequence impedances of this configuration are utilized in the positive-sequence reactance and Takagi methods. The results are shown in Table V for the estimated positive-sequence reactance as well as the fault distance. The error in the estimates is calculated using (3). It can be seen from Table V that the difference between the actual and the estimated distance for the positive-sequence reactance and Takagi methods is 0.15 and 0.14 miles, respectively. The corresponding absolute error is around -5% for both methods. This small value of error indicates that the proposed approach yields reasonably accurate fault location estimates for utility test case 1.

TABLE V
FAULT LOCATION ESTIMATES FOR UTILITY TEST CASE 1
Utility Test Case 1 –Actual Fault Distance is 2.67 miles

Method	Estimated Positive Seq. Reactance (Ω)	Estimated Distance (miles)	Absolute Difference (miles)	Error from Actual Location (%)
Positive-sequence Reactance	1.40	2.52	0.15	$\frac{2.52-2.67}{2.67} * 100 = -5.61$
Takagi	1.40	2.53	0.14	$\frac{2.53-2.67}{2.67} * 100 = -5.24$

B. Utility Test Case 2

The circuit diagram for utility test case 2 was constructed in OpenDSS [9]. This circuit is considered to be a high tech circuit by the utility. The circuit does not serve many customers, and there are no overhead transformers on the main line. The rated voltage at the substation is 23.9 kV, while the short-circuit capacity calculated using the circuit model is 239 MVA. A SEL-351S relay at the substation samples the three-phase line currents and the line-to-neutral voltages at a rate of 16 samples per cycle.

On 16th June 2010 at 08:45, several industrial customers called to complain about a fault on the circuit. The relay recorded voltage and current waveforms along with the fault log for this single line-to-ground fault on phase C are shown in Fig. 5. The relay recorded a prefault load current of about 200 A, while the maximum fault current magnitude recorded was 1.907 kA. The cause of the fault was found to be a squirrel contact with a lightning arrester located 5.33 miles from the substation.

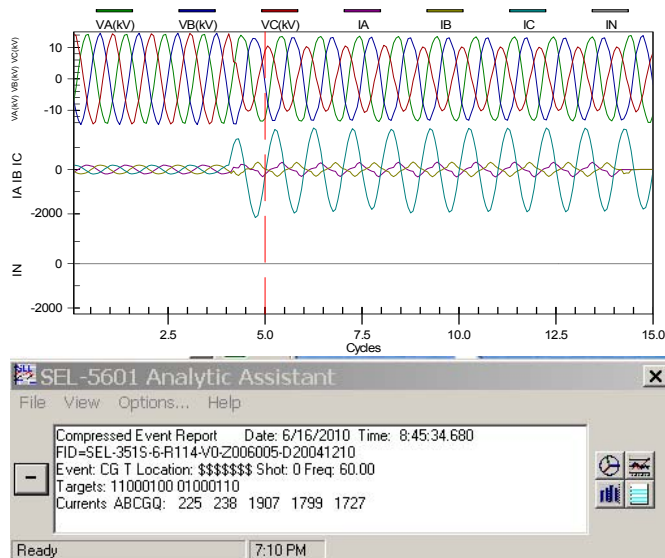


Fig. 5. Relay Data for Utility Test Case 2

The main feeder of this circuit is constructed using three different three-phase line configurations. The positive- and zero-sequence impedances and conductor lengths of these configurations are given in Table VI. The most commonly occurring conductor type in the circuit has the code, “556-AAC-HW,556-AAC-HW,556-AAC-HW,336-AAC”.

Using the proposed simplification the positive- and zero-sequence impedances of this configuration are utilized in the positive-sequence reactance and Takagi methods. The results are shown in Table VII for the estimated positive-sequence reactance as well as the fault distance. The error of the estimates is calculated using (3).

TABLE VI
LINE CONFIGURATION DATA FOR UTILITY TEST CASE 2

Parameter	2-ACSR, 2 ACSR, 2-ACSR, 2-ACSR	336-AAC, 336-AAC, 336-AAC, 1/0-ACSR	556-AAC-HW, 556-AAC-HW, 556-AAC-HW, 336-AAC
z_0 (Ω /mile)	2.0238 + j2.4838	0.5502 + j2.0681	0.4372 + j1.9305
z_1 (Ω /mile)	1.3682 + j0.7010	0.2781 + j0.5986	0.1680 + j0.6218
length (mile)	0.13	1.25	6.23
location	small segment just up-stream from fault	branch of main feeder near substation	bulk of main feeder from substation

TABLE VII
FAULT LOCATION ESTIMATES FOR UTILITY TEST CASE 2
Utility Test Case 2–Actual Fault Distance is 5.33 miles

Method	Estimated Positive Seq. Reactance (Ω)	Estimated Distance (miles)	Absolute Difference (miles)	Error from Actual Location (%)
Positive-sequence Reactance	2.97	4.78	0.55	$\frac{4.78-5.33}{5.33} * 100 = -10.32$
Takagi	3.02	4.86	0.47	$\frac{4.86-5.33}{5.33} * 100 = -8.82$

It can be seen from Table VII that the difference between the actual and the estimated distance for the positive-sequence reactance and Takagi methods is 0.55 and 0.47 miles, corresponding to an error of -10.32% and -8.82% respectively. The error percentage in this case is slightly more than that for test case 1. However, the error is still low and it again indicates that the proposed approach can be applied to non-homogenous feeders without significant loss of accuracy.

C. Summary of All Utility Test Cases

Eight more utility test cases with non-homogenous feeders were analyzed to demonstrate the proposed approach. The distance to the fault estimates are summarized in Table VIII for all ten cases. Additionally, Table VIII gives the value of the absolute difference between the estimated and the actual distance in miles as well as the error percentages calculated using (3). The average absolute error for the positive-sequence reactance and Takagi estimates is 10.23% and 9.34% respectively. On the other hand, the median value of the absolute difference between the estimate and the actual distance is 0.16 miles and 0.15 miles, for the positive-sequence reactance and Takagi methods, respectively. Furthermore, the minimum and maximum difference between the estimate and the actual distance for the positive-sequence reactance method is 0.1 and 0.55 miles, respectively, while these values are 0.01 and 0.47 miles for the Takagi method.

TABLE VIII
DISTANCE TO THE FAULT ESTIMATES FOR UTILITY TEST CASES

Test Case No.	Actual Distance to Fault	Positive-sequence Reactance Method			Takagi Method		
		Distance Estimate (miles)	Abs. Diff. (miles)	Error (%)	Distance Estimate (miles)	Abs. Diff. (mile)	Error (%)
1	2.67	2.52	0.15	-5.62	2.53	0.14	-5.24
2	5.33	4.78	0.55	-10.32	4.86	0.47	-8.82
3	2.67	2.54	0.13	-4.87	2.55	0.12	-4.49
4	2.67	2.51	0.16	-5.99	2.52	0.15	-5.62
5	2.67	2.49	0.18	-6.74	2.50	0.17	-6.37
6	3.10	3.31	0.21	6.77	3.43	0.33	10.65
7	1.70	1.58	0.12	-7.06	1.71	0.01	0.59
8	0.93	1.09	0.16	17.2	1.10	0.17	18.28
9	10.70	10.48	0.22	-2.06	10.83	0.13	1.21
10	0.28	0.18	0.1	-35.71	0.19	0.09	-32.14
Average Absolute Error (%)		10.23			9.34		

The objective of this paper was to demonstrate that the impedance-based fault locations methods can be effectively applied to non-homogenous feeders by using the line parameters of the most commonly occurring conductor type in the circuit. The low error values in Table VIII for the positive-sequence reactance and Takagi methods indicate the effectiveness of the proposed approach. Furthermore, for feeders with two or more conductor types of equal length, the distance to the fault should be calculated using the parameters of each conductor type in the positive-sequence reactance and Takagi methods. Then a mean of the fault distance estimates obtained by using each conductor type should be taken. The demonstration of fault location on such a feeder having more than one commonly occurring conductor type is a part of future work.

VII. CONCLUSION

The paper focused on two impedance-based fault location algorithms, namely, the positive sequence reactance and Takagi methods. Among all other assumptions made by these methods the most critical is that of line homogeneity. However, this assumption is violated in practical distribution circuits as they are constructed using different conductors throughout the length. The paper proposed an approach which can be employed to locate faults on non-homogenous distribution feeders using these impedance-based methods. It was proposed to assume the feeder to be constructed using the most commonly occurring conductor type in the circuit. This approach was tested on the modified IEEE 34 Node Test Feeder modeled in PSCAD by simulating two cases, namely, the reference case and the case with a homogenous feeder. For the positive-sequence reactance method the maximum absolute error in both cases was about 8%. On the other hand, the maximum absolute error for the Takagi method was 6.92% and 9.53% in the homogenous case and reference case, respectively. In addition, the proposed approach was demonstrated using ten fault cases on utility distribution feeders. The average absolute error obtained for the positive-

sequence reactance and Takagi estimates was 10.23% and 9.34%, respectively, which corresponds to a median error value of 0.16 miles and 0.15 miles in the location estimates. In this way, the efficacy of the proposed approach was successfully tested with a simulated test feeder as well as utility data.

VIII. ACKNOWLEDGEMENT

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X. BIOGRAPHIES

Saurabh Kulkarni received his B.S. from the University of Pune, India in 2007 and the M.S. from The Department of Electrical and Computer Engineering at The University of Texas at Austin, where he is currently working towards his doctorate. His research interests are in power quality and fault diagnosis.

Neeraj Karnik received his B.Tech. degree from College of Engineering, Pune, India in 2008. He is currently working towards his M.S degree from The University of Texas at Austin. His research interests are in power system protection.

Swagata Das received her B.S. from SRM University, Chennai, India in 2009. She is currently working towards her M.S degree from The University of Texas at Austin. Her research interests are in power quality for AC and DC systems.

Surya Santoso (M'96–SM'02) received the M.S.E. and Ph.D. degrees in electrical and computer engineering from the University of Texas at Austin in 1994 and 1996, respectively. From 1997 to 2003, he was a Consulting Engineer with Electrotek Concepts. Currently, he is an Associate Professor in the Department of Electrical and Computer Engineering at the same university. His research interests include power quality, power systems, and wind power.